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Keywords: Cognitive Bias, External validity, Reverse innovation, Evidence-based Medicine

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Abstract

In this randomized, blinded crossover experiment in a sample of 347 English clinicians, respondents reviewed the same four abstracts twice, one month apart, changing only the author affiliation between the two occasions. Mean within-individual difference in ratings for strength of the evidence, relevance, and likelihood of referral to a peer was obtained through fixed effects models. There was a significant overall impact of high-income country source on respondents' ratings of relevance (4.50, 3.16 to 5.83, p<0.001) and recommendation to a peer (3.05, 1.77 to 4.33, p<0.001). Overall impact of high-income country source on assessment of the strength of the evidence in the abstracts was positive but not quite statistically significant (1.35, -0.06 to 2.76, p=0.053). Changing the author affiliation of an abstract from a Low- to a High-income country affects respondents' views of that abstract, all things being equal. Biases such as these can have far-reaching implications for the diffusion of knowledge and innovations from Low-income countries.

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Introduction

Extensive research suggests that bias may be influential in publication and citation of research and published research articles may be evaluated differentially based on the perceived (or actual) characteristics of the author – such as sex, rank, place of work and country-of-origin. For example, article acceptance rates have been shown to be higher when first authors live in English speaking rich nations, than non-English speaking rich nations [1]. An author's affiliation from the United States (US) can increase their citation counts by 20%, and articles focusing on the US or Europe have been reported to have a greater citation frequency compared with articles that focus on developing countries [2]. The likelihood of non-US abstracts being accepted for an American Heart Association Scientific meeting was significantly higher when the abstracts were reviewed without the author affiliation included (1.81 vs 1.41 $p<0.001$) [3]. Other studies find that journals favor authors located in their own country [4, 5]. Lee (2013) refers to this as nationality bias [6]. It is possible that articles with a focus on the US may simply be better articles, however

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scientists may, equally, be paying more attention to US articles whilst ignoring equally good articles conducted in a different location – this has been called an 'Americanization' of science [7].

However, in order to isolate the impact of bias in the evaluation of research it is essential to control for both the reviewer of the research and the research itself. Four studies control for quality and type of research in their assessment of the impact of social bias [8-11]. Peters and Ceci (1982) found that previously published articles resubmitted to the same journal that published them were subsequently rejected when author affiliations were altered [9]. However, their study did not control for the reviewer of the research. Harris et al (2016) found that changing the author affiliation of an abstract from High to Low income country led to a reduced likelihood of recommendation to a peer [10]. However, this research included between-group differences only. Godlee et al's randomized controlled study [11] assessed blinding on the quality of peer review, controlling for the type of article being assessed but also only examined between-group variation. Borsuk et al (2009) found that reviewer gender did not effect the outcome of reviewed articles based on

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author gender, however again, they do not control for the individual reviewer of the research [8]. Bruce et al's systematic review [12] noted a scarcity of methodologically robust studies examining the quality of peer review observing, in particular, the challenge of experimentally assessing peer review without revealing the purpose of the study.

We are less interested in ensuring that two people agree with respect to their judgment over a manuscript, and more interested that the same person agrees with him/herself when confronted with the same manuscript save for changing the institution and country of the author and the research. Given the paucity of research that controls for both the reviewer and the quality and type of research being reviewed, we conducted a randomized, controlled and blinded cross-over study to assess the within-individual change in evaluation of research abstracts when the source is experimentally altered, in this case from a high- to a low-income country.

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Methods

Trial design

We address this important research gap by conducting a randomized, blinded cross-over experiment, where participants rated the same abstracts on two separate occasions, one month apart, with the source of these abstracts changing, without their knowledge, between high- and low-income countries.

Participants – eligibility

Inclusion criteria were to be medically qualified clinicians, of any specialty, living and practicing in England at the time of participation in the study.

Study settings

We used a panel provider through the Qualtrics survey platform to recruit respondents to the survey. Panels are curated lists of people interested in participating in social research online. Respondents that fully completed the survey in Wave 1 were then re-contacted for the Wave 2

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survey four weeks later with the Abstract sources that they had received in Wave 1 reversed (see Exhibit 1). Two reminder messages were sent to the unfinished participants in Wave 2. Respondents who completed the survey were given an incentive for their time after each wave in accordance with market rates for clinician reimbursement for online surveys. The survey was soft launched on 8th July 2016 and fully launched on the 15th July 2016.

Interventions

Abstracts were purposively selected from Cochrane Reviews to ensure that there was a high internal validity for the type of study that it is describing and likely to be of at least some interest to most clinicians [13-16]. All abstracts were of similar length and complexity. We also included one control abstract on the topic of mother-to-child transmission of HIV whose source (Oxford University, England) did not change at all between the two rounds to account for any within-individual variation in ratings over time.

Abstract sources, listed as 'Author Affiliation', were fictionalized for institution and country of origin. High-

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income source countries (United States and Germany) were selected from the top ten countries by Gross Domestic Product (GDP) per capita (>$36000 per capita), and Organization for Economic Cooperation and Development (OECD) membership. Low-income source countries (Ethiopia and Malawi) were selected from the bottom ten countries by GDP per capita (<$1046 per capita), using 2015 World Bank data (http://data.worldbank.org/indicator/NY.GDP.PCAP.CD). The institutional affiliation was fictionalized to one of the top-five universities that also have a medical or healthcare faculty, in the respective countries. These were Harvard University (US), Freiburg University (Germany), University of Addis Ababa (Ethiopia) and University of Mzuzu (Malawi). We used the 2014 Times Higher Education World rankings (http://www.timeshighereducation.co.uk/world-university-rankings/2014-15/world-ranking) for the high-income country sources, and the http://www.4icu.org website for international rankings of institutions for the low-income sources. We also included the name of fictionalized journals, listed as 'Journal' on the Abstract, to ascertain the relative effect of journal Impact Factor (IF) in the rating of the abstracts. The high-impact journal used was the New England Journal of Medicine (IF 47.05) and the low-

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impact journal was the Journal of Community Medicine and Health Education (IF 0.85).

Outcomes

Each abstract was followed by the same three questions. First, how strong is the evidence presented in this abstract? Second, how relevant to you is the research in the abstract? Third, how likely are you to recommend this abstract to a colleague? Responses were on a scale (0 to 100) with 0 as the least (i.e. not at all strong, not at all relevant, not at all likely) and 100 as the most (extremely strong, extremely relevant, extremely likely).

Sample size

Based on the effect size detected in our between-group US pilot study [10], we calculated that a sample size of 52 individuals completing both waves of evaluation is required to detect within-individual differences of 2 (scale 0-100) for each abstract for a power of 80% and type 1 error of 5%.

Randomisation

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After completion of screening questions to ensure fulfillment of inclusion criteria, respondents were randomized to groups A1 or B1 (for abstracts with NEJM as the journal type) or to C1 or D1 (for abstracts with JCM as the journal type) (See Exhibit 1), and invited to rate the five abstracts (four experimental and one control). To avoid a possible order effect, the order in which the abstracts were presented in the survey was randomized for each participant. The survey platform used simple randomization occurring in real time as the respondent enters the survey therefore the respondent is unaware that any randomization has taken place. The survey type (A1, B1, C1 or D1) that the respondent was randomized to in Wave 1 dictated the survey type he/she subsequently received in Wave 2 (i.e. A2, B2, C2 or D2) one month later.

Blinding

So that the purpose of the study did not influence the responses, the survey was described as a Speed Reading survey, and we requested respondents to complete the reading and rating of each abstract as quickly and carefully as possible, to enhance anchoring and fast-

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thinking [17]. The time taken to read and respond to each abstract was measured by the survey platform and this information was presented to each respondent upon completion of the survey to heighten the 'psychological realism' of the survey. We assessed the success of the blinding by asking respondents, at the conclusion of the second survey, whether they had noticed any changes in the survey design between the two waves. In each Wave, a mix of two Low-income and two High income country sources were used so the respondents were not likely to become aware of the purpose of the study when the sources were changed in Wave 2.

Analysis

Data were retrieved via Qualtrics in CSV format and analyzed using Stata/SE 13 (Statacorp, College Station, Texas). We used demographic covariates (age, sex, country of birth), and professional experience covariates (research exposure, peer review experience, educational attainment) to assess balance between the groups. Respondents' age was calculated based on a presumed mid-year birth and survey completion date of 31st January 2017. We first calculated the mean abstract ratings for each abstract, and for each

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question, and compared the mean within-individual difference in ratings between abstracts with high-income and low-income country sources using two-tailed t-tests. We also calculated the difference in ratings between the two survey blocks (i.e. Groups A and B, and Groups C and D), which gives an indication of the effect of journal type (high impact vs. low impact). We then estimated fixed-effects models for each abstract and outcome independently – as well as for all abstracts pooled - to account for unobserved heterogeneity at the respondent level and included a wave dummy to control for any trend in ratings between waves. We conducted sensitivity analyses in which we excluded the three respondents that had recognized that the source had changed between the two rounds.

Ethical considerations

Data were analyzed at the individual level, deidentified and aggregated to the sample population. The mild, non-harmful deception regarding the purpose of the study was necessary because awareness of the objective of the research will very likely bias responses. It impacted negligibly on the experience of the survey. The potential respondents were recruited from panel management companies

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that specialize in providing survey services and therefore the respondents have already agreed to participate in research surveys. Participation was remunerated in accordance with best practice guidance for the panel. The data were stored on password and firewall protected Imperial College computers accessible only by the researchers involved in the research. The Imperial College London Research Ethics Committee and Joint Research Compliance Officer approved the protocol for the research (ICREC 16IC3400).

Limitations

As an online survey a selection bias may be introduced, and this may affect the representativeness of the findings but only if participation in the survey is associated with bias against Low-Income country research. We addressed this issue by blinding the participants to the purpose of the study. The specific definition of strength of evidence, relevance and referral to a peer may be perceived differently by different participants, however we addressed this issue by examining within-individual variation in responses only. This study does not delineate the mechanism

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through which changing source impacts on individuals' reviews of the research abstracts and we cannot be certain whether these are due to unconscious or conscious biases. We used the high-low income country anchor to tease out potential applied biases in our respondents however it is beyond the scope of this article to examine the bias, or biases, that are playing out in the minds of our respondents, only to empirically measure the effect of them. Our research used abstracts and not full research articles and it is possible, although unlikely, that reviews might be different using full research articles. Forms of social bias will manifest and be reproduced at the point of consumption, whether the research is in abstract or long form.

[Insert Exhibit 1 here]

Results

We obtained 551 complete responses at baseline (Wave 1) and, of those, 63.0% completed the survey at follow-up approximately 30 days later (mean 28.3, 28.0 to 28.7) resulting in a longitudinal sample of 347 complete responses.

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Respondents were comparable within each group for a range of variables including gender, age, country of birth, years since qualification, percentage holding a doctoral degree, time spent in clinical practice, time spent reading the abstracts and experience of peer review.

Exhibit 2 shows that there was a significant overall impact of high-income country source on relevance (4.50, 3.16 to 5.83, $p<0.001$) and recommendation (3.05, 1.77 to 4.33, $p<0.001$). Perceived relevance was affected for three abstracts (Abstract 1: 2.69, 0.27 to 5.11, $p<0.05$; Abstract 2: 5.51, 3.01 to 8.02, $p<0.001$; Abstract 3: 8.09, 5.34 to 10.84, $p<0.001$). Likelihood of recommendation to a peer was affected for three abstracts (Abstract 1: 2.76, 0.27 to 5.26, $p<0.05$; Abstract 3: 5.50, 2.75 to 8.25, $p<0.001$; Abstract 4: 3.07, 0.54 to 5.60, $p<0.05$). The overall impact of high-income country source on assessment of the strength of the evidence in the Abstracts was positive but not quite statistically significant (1.35, -0.06 to 2.76, $p=0.053$), although a significant impact of high-income country source on strength of evidence was found for abstract 3 (3.98, 1.16 to 6.79, $p<0.001$). Findings were not altered when the three respondents (less than 1% of the

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sample) that noticed the change in source were excluded from the analysis.

There was no significant effect of the interaction between the journal type and the country source – the effect of country source was the same independent of the journal type. The effect of changing the country source was much more significant than the effect of journal type. In the between-group analysis (Exhibit 2) difference in ratings were insignificant for all abstracts except abstract 3 where the relevance of the abstract was rated lower with a higher-impact journal source (Abstract 3: -4.04, -8.16 to -0.08, $p<0.05$).

[Insert Exhibit 2 here]

Discussion

This study is, to the best of our knowledge, the first to measure the presence of explicit bias in abstract review controlling for both the reviewer and the abstract. We found that changing the source of an abstract from a Low to a High-income country led to a significant increase in the perceived relevance of the abstract and subsequent

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likelihood of referral of the abstracts. This finding was unaffected by whether the abstract was listed as having been published in a high- or a low-impact journal. The positive effect of changing the source from Low to a High-income country was significant for one abstract (an RCT) and also very nearly significant for rating the strength of the evidence of all the abstracts (1.35, -0.06 to 2.76, p=0.053)

For relevance and recommendation, ideally, we would not expect there to be any effect of the change in country sources, and yet for some of the abstracts, all things being equal, the effect was significant. The change in rating was up to 25% of the mean score for the abstract, in some instances. For the strength of evidence, there are well-developed and well-known criteria upon which this can be assessed, such as the hierarchy of evidence, where randomized, controlled trials and other experimental designs are preferred over observational designs and so we were not expecting the effect of changing source country to have any impact on this outcome. Yet, even this measure was affected positively and significantly, in one of the abstracts, a randomized controlled clinical trial, by the country source changing from Low to High-income.

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The marketing literature has known of the effect of country-of-origin on product evaluation for several decades [18-20] and the recruitment industry has made strides to ensure that candidates are treated fairly and equally by removing all identifiable information from curriculum vitae and job applications. We need to turn this attention to our review of medical research to avoid spurious admiration for research from some contexts to the detriment perhaps of other contexts, particularly when based on characteristics that are completely unrelated to the article's scientific merit.

The study touches on issues of external validity and how this is understood, measured and perceived by consumers of medical research. Whilst tools exist to measure the internal validity of a research study, there are very few tools to assess the external validity of that research. Whether research is considered relevant or not, ought not to be affected by where that research has come from given there are no accurate ways to assess how comparable two contexts are, or are not - if, indeed, relevance were to be based on context comparability at all. It follows, therefore, that any change in the perceived relevance of

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research, from one country compared to another, is due to unconscious (or conscious) cognitive biases, particularly if the research being judged is absolutely identical in every other respect – which it was in this study. Bias might not be conscious or malicious; it is simply a judgment as to whether results from a study will or wont apply in another context [21]. There are currently no approaches that effectively remove the possibility of bias in this judgment and preconceived notions of what does or does not constitute generalizable research will likely be heavily influenced by prior beliefs.

Some strategies do support the external validity of research such as increasing the sample size of the study, repeating it in a variety of contexts, describing the research context in enormous detail, such as through ethnography, so that arguments can be developed regarding similarity of settings. Steventon et al [22] describe a statistical modeling method to generalize findings from an RCT, but of course this is only a modeling technique. Cole and Stuart [23] describe how treatment effects from an RCT could be generalized by reweighting them to reflect the characteristics of patients in the target population. It is possible to embed RCTs within large routine data sets [24],

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increasing representativeness by sampling patients from multiple sites. Cochrane's DECIDE Framework and CERQUAL framework are useful to explore whether the generating mechanisms that explain intervention and outcome in the research context are present in the adopter context. However, none of these techniques are available to all, all of the time, in every research topic. Reporting guidelines, of which there are now over 300 (see www.equator-network.org) [25], will not necessarily prevent errors and biases in the interpretation of the report.

Based on the findings from this study, we expect that much research from low-income countries, even once it has passed through other publication barriers of peer review, has been and will continue to be discounted prematurely and unfairly, either through biased assessment of its rigor or its relevance. In previous research, we have found several barriers to the adoption of innovations from low-income countries, including frank prejudice [26], doubt as to whether contexts are similar enough to learn from [27] and assumed superiority in international health partnerships [28]. For healthcare workers in the US and UK, it is unusual for low-income countries to be considered sources of innovation [29]. So-called Reverse Innovation, the

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adoption of innovations from low-income countries into high-income contexts, has become increasingly studied, predominantly in the management and business literatures with a focus on the firm, but increasingly also of interest in the health policy space [30]. Identifying low-cost models of care that do not compromise on quality is a priority for many health systems. It is important to ensure that low-income countries are not discounted prematurely as a source of innovation, increasingly recognized as important features of international health partnerships and overseas volunteering [31] as well as a critical element of an innovation ecosystem [32].

Although there have been advances to protect against bias in the pre-publication stage, such as blinded peer review and open peer review, once the article is in the general domain there are no protections in place to ensure that the assessment of such research is free from bias.

Nationality bias is not necessarily the only bias potentially at work. Lee (2013) describes several other types, such as prestige bias, confirmation bias, low inter-rater reliability, affiliation bias, bias as a function of reviewer characteristics, content-based bias, confirmation

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bias and conservatism [6]. Merton and Zuckerman (1968) explain that the Matthew Effect is when the perceived reputation of the author influences the significance of the article [33]. It 'violates the norm of universalism embodied in the institution of science and curbs the advancement of knowledge' (33:62). Scientists are actually applying heuristics when conducting evaluation tasks, even if they are not aware (or admit to being aware) of them [34]. Several heuristics are potentially in play – the 'take-the-first heuristic', the 'fluency heuristic', the 'take-the-best heuristic', and the 'recognition heuristic' [21]. Whilst we consider the main distinguishing feature between Germany/Ethiopia and the US/Malawi to be the GDP per capita, it may well be that for some respondents other characteristics were being considered. The elicited bias is a complex function of the reviewer, the abstract content and the sources that were used. It is possible that the accentuated effect of source bias on Abstract 3 is related in part to the content of the abstract (a pharmaceutical RCT) or that the country sources for this abstract (Malawi and US) tend to elicit a greater bias. This could be investigated in further research.

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Conclusion

Our study is a significant contribution to the literature on peer reviewing because we control not only for the reviewer, the abstract and the order in which the abstracts were rated, but also we blind the reviewers to the purpose of the study.

One's viewpoint of a research article is a complex function of the context of the research, the research itself and the consumer of that research. We think there is merit in being more ambitious than publishing articles and hoping for the best. Options are to remove all author identifiable information from published articles and use meta-data instead to track citation rates and impact. Alternatively, to develop Risk of Bias tools, similar to the Cochrane Tools, not applied to the research, but to oneself. Either way, there is a clear need for further behavioral studies of evidence-based medicine including, but not limited to, factorial design studies to measure the impact of different sources, in different populations, using different types of research abstracts.

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References

1. Ross J, Gross C, Desai M, Hong Y, Grant A, Daniels S. Effect of Blinded Peer Review on Abstract Acceptance. *JAMA*. 2006; 295(14):1675-1680. doi:10.1001/jama.295.14.1675
2. Link, A.M (1998). US and non-US submissions. Journal of the American Medical Association, 280(3), 246-247
3. Daniel H-D. An Evaluation of the Peer Review process at Angewundte Cftemie. Angew. Chem. hi. Ed. Engl. 1993. 32, No. 2: 234-238
4. Ernst, E., & Kienbacher, T. (1991). Chauvinism. Nature, 352, 560
5. Tregenza T. Gender bias in the refereeing process? Trends in Ecology and Evolution 2002; 17(8):349-350

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6. Lee C, Sugimoto, Zhang G and Cronin B. Bias in Peer Review. Journal of the American Society for Information Science and Technology, 2013, 64(1):2-17.
7. FREY, B. S., R. EICHENBERGER (1993), American and European economics and economists, Journal of Economic Perspectives, 9: 185–194.
8. Borsuk, R.M., Aarssen, L.W., Budden, A.E., Koricheva, J., Leimu, R., Tregenza, T., Lortie, C.J., (2009). To name or not to name: The effect of changing author gender on peer review. Bioscience, 59(11), 985–989
9. Peters, D.P., & Ceci, S.J. (1982a). Peer-review practices of psychological journals: The fate of accepted, published articles, submitted again. Behavioral and Brain Sciences, 5(2), 187–195
10. Harris M, Macinko J, Jimenez G, Mahfoud M, Anderson C. Does the origin of research affect perception of research quality and relevance? A national trial of US public health academics. BMJ Open 2015;5:e008993. doi:10.1136/bmjopen-2015-008993
11. Godlee F, Gale CR, Martyn CN. Effect on the quality of peer review of blinding reviewers and asking them to sign their reports: a randomized controlled trial. JAMA 1998; 280(3):237-40

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12. Bruce R, Chauvin A, Trinquart L Ravaud P and Boutron I. Impact of interventions to improve the quality of peer review of biomedical journals: a systematic review and meta-analysis. BMC Medicine 2016 14;85

13. Clarke M, Dick J, Zwarenstein M, Lombard C, Diwan V. Lay health worker intervention with choice of DOT superior to standard TB care for farm dwellers in South Africa: a cluster randomised control trial. INT J TUBERC LUNG DIS 2005:9(6):673–679

14. Chabikuli NO, Awi DD, Chukwujeku O, Abubakar Z, Gwarzo U, Ibrahim M, Merrigan M, Hamelmann C. The use of routine monitoring and evaluation systems to assess a referral model of family planning and HIV service integration in Nigeria. AIDS 2009 Nov;23 Suppl 1:S97-S103.

15. Peters S, Palmer M, Grobbee D, Crouse J, O'Leary D, Raichlan J, Bots M. C-reactive protein lowering with rosuvastatin in the METEOR study. Journal of Internal Medicine 2010: Volume 268, Issue 2; p 155–161

16. Mark H, Nanda J, Davis-Vogel A, Navaline H, Scotti R, Wickrema R et al. Profiles of self-reported HIV risk behaviours among injection drug users in methadone maintenance treatment, detoxification and

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needle exchange programs. Public Health Nursing 2006; 23(1): 11-19

17. Tversky, A and Kahneman, D. Judgment under uncertainty: Heuristics and biases. *Science*, 185 (1974), 1124-1131.

18. Dinnie K. Country-of-origin 1965-2004: a systematic literature review. Journal of Customer Behaviour.

19. Bilkey, W. and Nes, E. Country-of-Origin Effects on Product Evaluations. *Journal of International Business Studies*. 1982;Vol. 13, Issue 1, pp. 89-100, 1982

20. Peterson, Robert A. and Jolibert A. A Meta-Analysis of Country-of-Origin Effects. *Journal of International Business Studies*. 1995; 26 (4), 883-900

21. Kukal W and Ganguli M – Generalizability. Neurology 2012; 78: 1886-1891

22. Steventon, A, Grieve R, Martin Bardsley, PhD An Approach to Assess Generalizability in Comparative Effectiveness Research: A Case Study of the Whole Systems Demonstrator Cluster Randomized Trial Comparing Telehealth with Usual Care for Patients with Chronic Health Conditions. Medical Decision Making 2015 DOI: 10.1177/0272989X15585131

For Review Only

23. Cole SR, Stuart EA. Generalizing evidence from randomized clinical trials to target populations: the ACTG 320 trial. Am J Epidemiol. 2010;172(1):107–15.

24. Van Staa T-P, Goldacre B, Gulliford M, et al. Pragmatic randomized trials using routine electronic health records: putting them to the test. BMJ. 2012;344:e55.

25. Rennie D. Make Peer Review Scientific. Nature. July 2016 Vol 35; 31-33

26. Johnson C, Noyes J, Haines A, Thomas K, Stockport C and Harris M. Community Health Workers – Learning from the Brazilian Model in North Wales. Globalization and Health 2013, 9:25 doi:10.1186/1744-8603-9-25

27. Harris M, Weisberger E, Macinko J. 'They hear "Africa" and think there are no good services there' - perceived context in cross-national learning: a qualitative study. Globalization and Health 2015: 11;45

28. Kulabasanthan K, Issa, H, Bhatti Y, Prime M, Quinn M, del Castillo J et al. Do International Health Partnerships contribute to Reverse Innovation? a mixed methods study of THET-supported partnerships. Globalization and Health 2017: 13;25

For Review Only

29. Harris M, Bhatti Y, Prime M, del Castillo J, Parston G, Darzi A. Global Diffusion of Healthcare Innovation: Making the Connections. Doha, Qatar: World Innovation Summit for Health, 2016

30. Hadengue M, de Marcellis-Warin N, Warin T. Reverse Innovation A Systematic Literature Review. International Journal of Emerging Markets, Vol. 12(2) pp.1-25

31. Tropical Health Education Trust. In Our Mutual Interest. April 2017. Available from http://www.thet.org/resource-library/in-our-mutual-interest

32. Heitmueller A, Bull A, Oh S. Looking in the wrong places: why traditional solutions to the diffusion of innovation will not work. BMJ Innovations 25 March 2016

33. Merton RK. The Matthew Effect in Science. Science 1968 Vol 159:56-63

34. Bornmann, L. (2015a). Complex tasks and simple solutions: the use of heuristics in the evaluation of research. *Journal of the Association for Information Science and Technology, 66*(8), 1738-1739

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Exhibit List

1. Exhibit 1 (Table) – Caption: Sources attributed to abstracts, and respondent characteristics.
2. Exhibit 2 (Table) - Caption: Ratings, according to stated source, by research abstract

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Table 1 – Sources[a] attributed to abstracts, and respondent characteristics.

Group[b]	A	B	C	D
Sample size	81	89	89	88
Journal attribution (all abstracts)	NEJM	NEJM	JCM	JCM
University/Country attributions				
Control Abstract [c] Antenatal care quality				
Wave 1	University of Oxford/UK	University of Oxford/UK	University of Oxford/UK	University of Oxford/UK
Wave 2	University of Oxford/UK	University of Oxford/UK	University of Oxford/UK	University of Oxford/UK
Abstract 1 DOTS TB randomised controlled trial				
Wave 1	University of Freiburg/Germany	University of Addis/Ethiopia	University of Freiburg/Germany	University of Addis/Ethiopia
Wave 2	University of Addis/Ethiopia	University of Freiburg/Germany	University of Addis/Ethiopia	University of Freiburg/Germany
Abstract 2 Cross-sectional study, HIV care				
Wave 1	University of Addis/Ethiopia	University of Freiburg/Germany	University of Addis/Ethiopia	University of Freiburg/Germany
Wave 2	University of Freiburg/Germany	University of Addis/Ethiopia	University of Freiburg/Germany	University of Addis/Ethiopia
Abstract 3 Rosuvastatin randomised trial				
Wave 1	University of Muzuzu/Malawi	Harvard University/US	University of Muzuzu/Malawi	Harvard University/US
Wave 2	Harvard University/US	University of Muzuzu/Malawi	Harvard University/US	University of Muzuzu/Malawi
Abstract 4 Cross-sectional study, methadone				
Wave 1	Harvard University/US	University of Muzuzu/Malawi	Harvard University/US	University of Muzuzu/Malawi
	University of	Harvard	University of	Harvard

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Wave 2	Muzuzu/Malawi	University/US	Muzuzu/Malawi	University/US
Respondent characteristics (95% CI)				
Males, %	69.1 (57.9–78.9)	65.2 (54.3–74.0)	71.9 (61.4–80.9)	79.3 (62.3–87.3)
Mean age, years	45.5 (43.8–47.3)	45.5 (43.7–47.2)	44.1 (42.2–45.9)	43.5 (41.6–45.5)
Born in the UK, %	55.6 (44.1–66.6)	66.3 (55.5–76.0)	62.9 (52.0–72.9)	69.3 (58.6–78.7)
Doctoral degree, %	14.8 (7.9–24.4)	12.4 (6.3–21.0)	19.1 (11.5–28.8)	26.1 (17.3–36.6)
Frequent consumer of research, %[a]	33.3 (23.2–44.7)	32.6 (23.0–43.3)	39.3 (29.1–50.3)	38.6 (28.4–49.6)
Frequent peer reviewer of research, %[b]	4.90 (1.4–12.2)	1.12 (0.01–6.1)	6.70 (2.5–14.1)	6.80 (2.5–14.3)
Mean time since qualification, years	21.0 (19.4–22.8)	21.4 (19.5–23.3)	20.3 (18.3–22.4)	19.3 (17.4–21.2)
Proportion spending 3 or more days in clinical practice per week, %	95.1 (87.8–98.6)	97.8 (92.1–99.7)	93.3 (85.9–97.5)	94.3 (87.2–98.1)
Survey response				
Mean time spent per Abstract, seconds	83.0 (66.0–99.9)	71.7 (61.8–81.7)	76.4 (58.4–94.5)	70.0 (58.7–81.2)
Mean time between Wave 1 and 2, days	27.9 (27.4–28.5)	28.1 (27.5–28.7)	28.9 (28.3–29.6)	28.3 (27.7–29.0)

NEJM – New England Journal of Medicine; JCM – Journal of Community Medicine and Health Education; [a] Based on GDP/capita using 2015 World Bank data (http://data.worldbank.org/indicator/NY.GDP.PCAP.CD). High-income countries selected from top ten countries and OECD membership and Low-income countries selected from the bottom ten countries by GDP per capita. [b] Respondents randomized to Groups A–D to control for a possible order effect. [c] Control abstract source remained unchanged between Waves to control for between-wave variation in rating.

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Table 2: Ratings, according to stated source, by research abstract (n=347)

	Low income country	High income country	Difference in ratings for High vs Low income country[a] (95% CI)	High Impact factor journal (IF 47.05)	Low impact factor journal (IF=0.85)	Difference in ratings for High vs Low impact factor journal[b] (95% CI)
Abstract 1: Randomised trial of DOTS treatment for tuberculosis						
Strength-mean (SD)	50.9 (22.8)	50.3 (23.3)	-0.54 (-3.32, 2.25)	50.2 (23.3)	51.0 (22.8)	-0.79 (-4.22, 2.64)
Relevance-mean (SD)	26.4 (24.7)	29.1 (25.9)	**2.69 (0.27, 5.11)***	26.0 (25.3)	29.4 (25.3)	-3.42 (-7.19, 0.35)
Recommendation-mean (SD)	27.3 (25.2)	29.9 (26.5)	**2.76 (0.27, 5.26)***	28.1 (26.1)	29.1 (25.7)	-0.94 (-4.79, 2.92)
Abstract 2: Cross-sectional comparison of HIV services in maternal and child health						
Strength-mean (SD)	42.9 (22.0)	44.9 (22.0)	1.97 (-0.52, 4.46)	43.9 (22.0)	43.9 (22.1)	-0.04 (-3.32, 3.25)
Relevance-mean (SD)	24.3 (24.6)	29.7 (26.1)	**5.51 (3.01, 8.02)*****	27.7 (27.4)	26.3 (23.4)	1.43 (-2.37, 5.23)
Recommendation-mean (SD)	26.3 (25.4)	27.6 (24.9)	1.30 (-1.11, 3.70)	27.7 (26.5)	26.1 (23.6)	1.6 (-2.16, 5.32)
Abstract 3: Randomised trial of cholesterol-lowering drug, Rosuvastatin						
Strength-mean (SD)	55.8 (23.7)	59.8 (23.0)	**3.98 (1.16, 6.79)*****	57.3 (23.4)	58.3 (23.5)	-1.0 (-4.49, 2.50)
Relevance-mean (SD)	34.4 (27.1)	42.4 (27.8)	**8.09 (5.34, 10.84)*****	36.3 (28.5)	40.3 (26.7)	**-4.04* (-8.16, -0.08)**
Recommendation-mean (SD)	33.6 (27.2)	39.0 (27.3)	**5.50 (2.75, 8.25)*****	34.4 (27.8)	38.0 (26.9)	-3.6 (-7.68, 0.48)
Abstract 4: Cross-sectional trial of						

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methadone treatment in drug addicts						
Strength–mean (SD)	39.5 (23.8)	39.5 (22.0)	0.10 (-2.63, 2.83)	38.9 (23.5)	40.1 (22.4)	-1.25 (-4.67, 2.17)
Relevance–mean (SD)	25.8 (23.8)	27.9 (24.5)	2.00 (-0.48, 4.47)	25.6 (23.7)	28.1 (24.6)	-2.45 (-6.05, 1.15)
Recommendation–mean (SD)	24.5 (23.3)	27.6 (24.9)	**3.07 (0.54, 5.60)***	26.1 (23.7)	26.0 (24.6)	0.04 (-3.55, 3.64)
Combined results across abstracts						
Strength–mean (SD)	47.3 (23.8)	49.4 (23.7)	1.35 (-0.06, 2.76)	48.4 (24.2)	48.7 (23.4)	-0.30 (-1.88, 1.29)
Relevance–mean (SD)	27.7 (25.4)	33.4 (27.3)	**4.50 (3.16, 5.83)*****	30.4 (27.3)	31.8 (26.0)	-1.44 (-3.21, 0.34)
Recommendation–mean (SD)	27.9 (25.5)	32.1 (26.8)	**3.05 (1.77, 4.33)*****	30.3 (26.8)	30.5 (26.0)	-0.28 (-2.03, 1.48)

*p<0.05; **p<0.01; ***p<0.001
IF = impact factor
[a] fixed effects model, to account for unobserved heterogeneity at respondent level, and adjusted for wave
[b] unadjusted between-group comparisons

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Supplementary files

Informed Consent

Informed Consent Form

Introduction

This survey explores speed reading and examines whether the amount of time that you spend reading a research abstract affects how you rate it.

Procedures

You will be shown five different healthcare research abstracts for you to read as quickly but as carefully as you can. Each abstract is on a different topic but broadly within the subject area of health and healthcare. After each abstract you will be asked the same three questions that you should respond to. Questions are designed to find out what you thought of the abstract - there is no right or wrong answer. Once you complete the survey, we will invite you to repeat the survey a second time about three weeks later. Taking the survey twice will allow us to compare how consistent you are in your responses.

Risks/Discomforts

The survey will take about ten minutes to complete each time. Risks are minimal for involvement in this study.

Benefits

Through your participation, researchers will learn more about how speed reading affects peoples' interpretation of research evidence.

Confidentiality

All data obtained from participants are anonymous and confidential and will only be reported in an aggregate format (by reporting only combined results and never reporting individual ones). The data collected will only be used for this research and other approved analyses. The data will be stored in the HIPPA-compliant, Qualtrics-secure database until it has been deleted by the primary investigator.

Compensation

You will be compensated for your involvement in this survey. You will receive 50% for completing the survey once, but full compensation if you complete the second survey in three weeks time.

Participation

Participation in this research study is completely voluntary. You have the right to withdraw at anytime or refuse to participate entirely without jeopardy.

Questions about the Research

The research protocol has been reviewed by the Imperial College London Joint Research Compliance Office. If you have questions regarding this study, you may contact the survey panel provider that invited you to take this survey.

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Abstracts used in the study

Control Abstract

Quality of antenatal and delivery care before and after the implementation of a prevention of mother-to-child HIV transmission programme, in the United Kingdom.

AUTHOR AFFILIATION:

University of Oxford, United Kingdom

JOURNAL:

The New England Journal of Medicine

OBJECTIVE:

To assess whether implementation of a prevention of mother-to-child HIV transmission (PMTCT) programme improved the quality of antenatal and delivery care services, in the United Kingdom.

METHODS:

Quality of antenatal and delivery care services was assessed in five urban health facilities before (2002-2003) and after (2005) the implementation of a PMTCT programme through review of facility data; observation of antenatal consultations (n = 606 before; n = 591 after) and deliveries (n = 229 before; n = 231 after) and exit interviews of women; and interviews of health facility staff.

RESULTS:

HIV testing was never proposed at baseline and was proposed to 63% of women at the first antenatal care visit after PMTCT implementation. The overall testing rate was 42% and 83% of tested HIV-infected pregnant women received nevirapine. In addition, inter-personal communication and confidentiality significantly improved in all health facilities. In the maternity ward, quality of obstetrical care at admission, delivery and post-partum care globally improved in all facilities after the implementation of the programme although some indicators remained poor, such as filling in the partograph directly during labour. Episiotomy rates among primiparous women dropped from 64% to 25% ($P < 0.001$) after PMTCT implementation. Global scores for quality of antenatal and delivery care significantly improved in all facilities after the implementation of the programme.

CONCLUSIONS:

In the United Kingdom, introducing comprehensive PMTCT services can improve the quality of antenatal and delivery care in general.

Abstract 1

Lay health worker intervention with choice of DOT superior to standard TB care for farm dwellers in Germany: a cluster randomised control trial

AUTHOR AFFILIATION:

University of Freiburg, Germany

JOURNAL:

The New England Journal of Medicine

OBJECTIVE:

To evaluate the effect of lay health workers (LHWs) on tuberculosis (TB) control among permanent farm workers and farm dwellers, in Germany, in an area with particularly high TB prevalence.

DESIGN:

Pragmatic, unblinded cluster randomised control trial.

METHODS:

This trial measured successful treatment completion rates among new smear-positive (NSP) adult TB patients on 106 intervention farms, and compared them with outcomes in patients on 105 control farms. Farms were the unit of randomisation, and analysis was by intention to treat.

RESULTS:

A total of 164 adult TB patients were recruited into the study, 89 of whom were NSP. The successful treatment completion rate in NSP adult TB patients was 18.7% higher (P = 0.042, 95%CI 0.9-36.4) on farms in the intervention group than on farms in the control group. Case finding for adult NSP TB cases was 8% higher (P = 0.2671) on farms in the intervention group compared to the control group.

CONCLUSION:

In Germany, trained LHWs were able to improve the successful TB treatment rate among adult NSP TB patients in a well-established health service, despite reduction of services.

Abstract 2

The use of routine monitoring and evaluation systems to assess a referral model of family planning and HIV service integration in Ethiopia.

AUTHOR AFFILIATION:

University of Addis Ababa, Ethiopia

JOURNAL:

The New England Journal of Medicine

OBJECTIVE:

To measure changes in service utilization of a model integrating family planning with HIV counselling and testing (HCT), antiretroviral therapy (ART) and prevention of mother-to-child transmission (PMTCT) in the Ethiopian public health facilities.

DESIGN:

It is a retrospective survey of attendance and family planning commodity uptake in 71 health facilities in Ethiopia that analyzes the preintegration and postintegration periods between March 2007 and January 2009.

METHODS:

A prepost retrospective comparison of mean attendance at family planning clinics and couple-years of protection (CYP) compared 6 months preintegration with 9 months postintegration period. An analysis of service ratios was conducted, relating completed referrals at family planning clinics to service utilization at the referring HIV clinics.

RESULTS:

Mean attendance at family planning clinics increased significantly from 67.6 in preintegration to 87.0 in postintegration. The mean CYP increased significantly from 32.3 preintegration to 38.2 postintegration. Service ratio of referrals from each of the HIV clinics was low but increased in the postintegration period by 4, 34 and 42 per 1000 clients from HCT, ART and PMTCT clinics, respectively. Service ratios were higher in primary healthcare settings than in secondary or tertiary hospitals. Attendance by men at family planning clinics was significantly higher among clients referred from HIV clinics.

CONCLUSION:

Family planning-HIV integration using the referral model improved family planning service utilization by clients accessing HIV services, but further improvement is possible. Male utilization of family planning services also improved. The government of Ethiopia should review the family planning user fee policy and scale up the integration in primary healthcare facilities.

Abstract 3

C-reactive protein lowering with rosuvastatin in Malawi.

AUTHOR AFFILIATION:

University of Muzuzu, Malawi

JOURNAL:

The New England Journal of Medicine

OBJECTIVES:

In addition to its LDL-C-lowering effects, statin treatment reduces the level of C-reactive protein (CRP). Long-term data on this effect in low-risk populations are limited. Furthermore, whether the CRP reduction is a consequence of LDL-C lowering or occurs independently remains unclear. We studied these aspects in a randomized placebo-controlled trial amongst 984 low-risk subjects, in Malawi.

METHODS:

A randomized placebo-controlled trial that evaluated the effect of 40 mg of rosuvastatin on 2-year change in carotid intima media thickness (CIMT) amongst 984 low-risk patients (10-year Framingham risk < 10%) with modest CIMT (CIMT > or = 1.2 and < 3.5 mm) and elevated LDL-C. CRP levels were measured at baseline and after 2 years of treatment.

RESULTS:

Median baseline CRP was 1.4 mg L(-1). Rosuvastatin lowered CRP significantly compared with placebo: -36% in the rosuvastatin group versus no change in the placebo group. There was no relation between change in CRP and change in LDL-C (Spearman correlation: 0.08; SE: 0.04). Stratified analyses showed that the CRP-lowering effect was present amongst all strata of baseline characteristics, including baseline lipids and CRP levels. However, the magnitude of CRP reduction was larger amongst women and participants older than 60 years.

CONCLUSIONS:

In Malawi, Rosuvastatin (40 mg) lowers CRP independently from its effects on LDL-C in low-risk subjects with normal baseline CRP levels and modest CIMT.

Abstract 4

Profiles of self-reported HIV-risk behaviors among injection drug users in methadone maintenance treatment, detoxification, and needle exchange programs, in the United States.

AUTHOR AFFILIATION:

Harvard University, United States

JOURNAL:

The New England Journal of Medicine

OBJECTIVE:

Injection drug use has accounted for more than one third of acquired immune deficiency syndrome cases in the United States. The purpose of this study was to compare the demographic characteristics, types, and frequency of human immunodeficiency virus (HIV)-risk behaviors among injection drug users (IDUs) recruited from a needle exchange program (NEP), methadone maintenance treatment (MMT), and detoxification (detox) program.

DESIGN:

A cross-sectional, correlational design was used to determine whether the selected HIV-risk behaviors and demographic characteristics of IDUs varied by site of recruitment.

SAMPLE AND MEASUREMENTS:

Confidential questionnaires were completed by 445 IDUs in the United States.

RESULTS:

Data analysis revealed that HIV sexual and injection-risk behavior varied by recruitment site. Subjects recruited from the NEP were more likely to engage in HIV-risk behaviors than subjects recruited from the MMT or detox sites.

CONCLUSIONS:

In the United States, interventions occurring in program and treatment sites need to be sensitive to various demographic characteristics and behaviors if they are to reach those at highest risk of HIV infection. Targeting HIV prevention interventions based upon risk group membership alone (e.g. IDUs) fails to address the distinct risk behaviors and demographic characteristics of enrollees in different programs.

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Example of Abstract with Low-income and High-income sources

Abstract 3: Low-income country source

C-reactive protein lowering with rosuvastatin in Malawi.

AUTHOR AFFILIATION:

University of Muzuzu, Malawi

JOURNAL:

The New England Journal of Medicine

OBJECTIVES:

In addition to its LDL-C-lowering effects, statin treatment reduces the level of C-reactive protein (CRP). Long-term data on this effect in low-risk populations are limited. Furthermore, whether the CRP reduction is a consequence of LDL-C lowering or occurs independently remains unclear. We studied these aspects in a randomized placebo-controlled trial amongst 984 low-risk subjects, in Malawi.

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CONCLUSIONS:

In Malawi, Rosuvastatin (40 mg) lowers CRP independently from its effects on LDL-C in low-risk subjects with normal baseline CRP levels and modest CIMT.

Abstract 3: High-income country source

C-reactive protein lowering with rosuvastatin in the United States

AUTHOR AFFILIATION:

Harvard University, United States

JOURNAL:

The New England Journal of Medicine

OBJECTIVES:

In addition to its LDL-C-lowering effects, statin treatment reduces the level of C-reactive protein (CRP). Long-term data on this effect in low-risk populations are limited. Furthermore, whether the CRP reduction is a consequence of LDL-C lowering or occurs independently remains unclear. We studied these aspects in a randomized placebo-controlled trial amongst 984 low-risk subjects in the United States.

METHODS:

A randomized placebo-controlled trial that evaluated the effect of 40 mg of rosuvastatin on 2-year change in carotid intima media thickness (CIMT) amongst 984 low-risk patients (10-year Framingham risk < 10%) with modest CIMT (CIMT > or = 1.2 and < 3.5 mm) and elevated LDL-C. CRP levels were measured at baseline and after 2 years of treatment.

RESULTS:

Median baseline CRP was 1.4 mg L(-1). Rosuvastatin lowered CRP significantly compared with placebo: -36% in the rosuvastatin group versus no change in the placebo group. There was no relation between change in CRP and change in LDL-C (Spearman correlation: 0.08; SE: 0.04). Stratified analyses showed that the CRP-lowering effect was present amongst all strata of baseline characteristics, including baseline lipids and CRP levels. However, the magnitude of CRP reduction was larger amongst women and participants older than 60 years.

CONCLUSIONS:

In the United States, Rosuvastatin (40 mg) lowers CRP independently from its effects on LDL-C in low-risk subjects with normal baseline CRP levels and modest CIMT.

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